

Similarly, the circuits are the columns of the matrices we describe next, and their negatives:

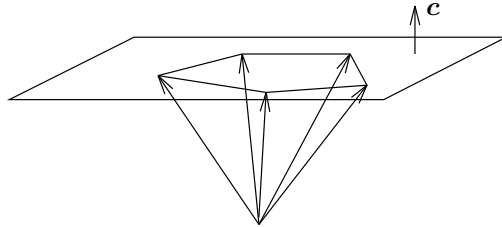
$$\mathcal{C}(X_1) : \begin{pmatrix} + & + & 0 \\ + & + & 0 \\ - & 0 & + \\ - & 0 & + \\ 0 & - & - \\ 0 & - & - \end{pmatrix} \quad \mathcal{C}(X_2) : \begin{pmatrix} + & + & + & 0 \\ + & + & + & 0 \\ - & - & 0 & + \\ - & 0 & + & + \\ 0 & - & - & - \\ 0 & - & - & - \end{pmatrix}.$$

6.2 Vector Configurations

While our discussion in Section 6.1 was on affine point configurations in $\mathbb{R}^d$, we now proceed to linear configurations of vectors in $\mathbb{R}^{d+1}$. The transition is the obvious one: with any configuration of points $\mathbf{x}_i$ in $\mathbb{R}^d$ (such as the vertices of a d -polytope), we associate the vectors $\mathbf{v}_i := \begin{pmatrix} 1 \\ \mathbf{x}_i \end{pmatrix}$ in $\mathbb{R}^{d+1}$. To get our notation for dimensions straight, we introduce a new parameter, called *rank*, as $r := d + 1$. As so often when dealing with a transition from affine to linear, it is convenient to have an extra letter r for affine rank (i.e., linear dimension), which is one more than the affine dimension d . Thus we have vectors $\mathbf{v}_i \in \mathbb{R}^r$.

In fact, what we get this way is an *acyclic vector configuration* $V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subseteq \mathbb{R}^r$, characterized by the following two properties (which are equivalent by a simple application of Farkas lemma II):

- (i) There is no nonnegative dependence, i.e., no $\mathbf{y} \geq \mathbf{0}$, $\mathbf{y} \neq \mathbf{0}$, such that $V\mathbf{y} = \mathbf{0}$.
- (ii) There is a linear function $\mathbf{c} \in (\mathbb{R}^r)^*$ such that $\mathbf{c}V > \mathbf{0}$ (i.e., $\mathbf{c}\mathbf{v}_i > 0$ for all i).



The construction of the sets of signed circuits, vectors, cocircuits, and covectors works for general vector configurations, acyclic or not. In fact, the linear algebra becomes slightly simpler in the linear case.

For this, let $V \in \mathbb{R}^{r \times n}$ be the matrix of a set of n vectors in $\mathbb{R}^r$. There may be multiple vectors, but we assume that the vectors span $\mathbb{R}^r$, so that $\text{rank}(V) = r$.

The space of *linear dependences* of the vector configuration V is

$$\text{Dep}(V) := \{\mathbf{v} \in \mathbb{R}^n : V\mathbf{v} = \mathbf{0}\} \subseteq \mathbb{R}^n.$$

This is a linear subspace of $\mathbb{R}^n$ of dimension $n - r$. The *signed vectors* of V are given by

$$\mathcal{V}(V) := \{\text{sign}(\mathbf{v}) \in \{+, -, 0\}^n : \mathbf{v} \in \mathbb{R}^n, V\mathbf{v} = \mathbf{0}\} = \text{SIGN}(\text{Dep}(V)),$$

and the *signed circuits* are the signed vectors of minimal nonempty support.

Dually, the space of *value vectors* on a vector configuration, which correspond to linear functions $\mathbf{c} \in (\mathbb{R}^r)^*$, is constructed as

$$\text{Val}(V) := \{\mathbf{c}V : \mathbf{c} \in (\mathbb{R}^r)^*\} \subseteq (\mathbb{R}^n)^*.$$

This is a linear subspace of $(\mathbb{R}^n)^*$ of dimension r . From it, we derive the set of *signed covectors* of the configuration V :

$$\mathcal{V}^*(V) := \{\text{sign}(\mathbf{c}V) : \mathbf{c} \in (\mathbb{R}^r)^*\} = \text{SIGN}(\text{Val}(V)).$$

Here we get the *signed cocircuits* as the signed covectors of minimal nonempty support: they correspond to the linear functions such that the vectors $\mathbf{v} \in V$ which have value 0 linearly span a hyperplane in $\mathbb{R}^r$.

Note that these definitions are consistent with our conventions for the affine case; see Exercise 6.3.

Proposition 6.4. *Let $V \in \mathbb{R}^{r \times n}$ represent a spanning configuration of n vectors in $\mathbb{R}^r$.*

Then $\text{Val}(V)$ is the set of all linear functions that vanish on all the vectors in $\text{Dep}(V)$, and $\text{Dep}(V)$ is the set of all vectors on which all the functions in $\text{Val}(V)$ vanish.

Proof. This follows from the dimension counts $\dim(\text{Dep}(V)) = n - r$, $\dim(\text{Val}(V)) = r$, with the computation $(\mathbf{c}V)\mathbf{v} = \mathbf{c}(V\mathbf{v}) = \mathbf{c}\mathbf{0} = 0$. $\square$

(If we identify $\mathbb{R}^n$ and $(\mathbb{R}^n)^*$ via the standard scalar product, then by the same argument we get that $\text{Dep}(V)$ is the orthogonal complement of $\text{Val}(V)$ in $\mathbb{R}^n$.)

6.3 Oriented Matroids

In this section, we meet oriented matroids for the first time. Just to keep things more exciting (and to make sure that the timid reader isn't overwhelmed by the first sight), we won't lift all the veils on the first encounter. So we won't even define oriented matroids this time, but we promise more for later.